ARIMA, VAR & Multivariate Models

ECON 8310: Business Forecasting — Lecture 2

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Lecture Outline

1 Stationarity and Differencing

2 ARIMA Models

3 VAR Models

4 Key Takeaways and Roadmap

Stationarity and Differencing

Before fitting any model, the series must have a stable mean and variance.

What Is Stationarity?

Weak (Covariance) Stationarity

A time series $\{y_t\}$ is **weakly stationary** if:

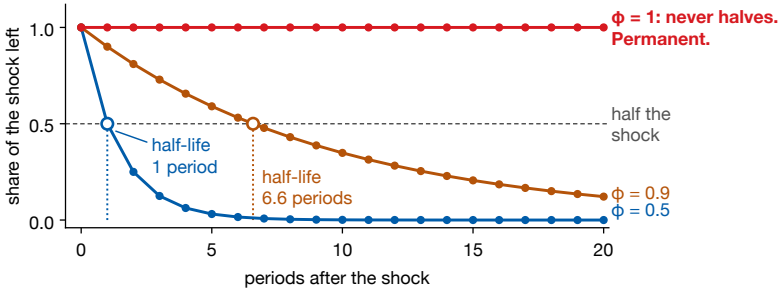
1. $\mathbb{E}[y_t] = \mu$ (constant mean over time)
2. $\mathrm{Var}(y_t) = \sigma^2 < \infty$ (constant, bounded variance)
3. $\mathrm{Cov}(y_t, y_{t-k}) = \gamma_k$ depends only on lag k, not on t

Why it matters: a drifting mean leaves no single value to forecast, and a growing variance widens prediction intervals without bound. Every model in this lecture assumes it.

The usual culprits: a trend, and variance that grows with the level. Differencing removes a stochastic trend; a log stabilizes the variance.

Unit Roots and Differencing

AR(1): $y_t = \phi_1 \, y_{t-1} + \varepsilon_t$. Give it one unit shock and watch what is left of it:



Below one, each period keeps the same share ϕ_1, so the shock halves every $\ln 0.5 / \ln \phi_1$ periods. At $\phi_1 = 1$, a **unit root**, it never fades: a random walk. **Differencing**, $\Delta y_t = y_t - y_{t-1}$, removes it. d counts first differences; D counts seasonal ones, $\Delta_{12} y_t = y_t - y_{t-12}$.

Testing for a Unit Root

Rule of thumb: a visible trend or growing variance means the series is likely non-stationary. Then test formally.

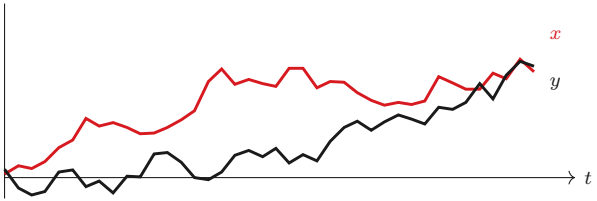
ADF (H_0: unit root) and **KPSS** (H_0: stationary) test *opposite* nulls, so run both, and be suspicious when they disagree.

> **Do not over-difference.** Differencing an already-stationary series induces negative autocorrelation that you then have to model away. Difference, re-test, and stop at $d = 1$ once the result is stationary.

Lecture 1's TX_1 HOUSEHOLD: ADF -1.38 on the levels (a unit root), -8.51 after one difference (stationary). So $d = 1$.

Why the Rule Exists: Spurious Regression

Differencing is not bookkeeping. Skip it and the statistics actively lie to you.



Two independent random walks — **no shared cause, no relationship at all**. Their *changes* correlate at -0.13, which is the truth. But their *levels* correlate at $+0.62$, and that is what OLS sees.

> Granger and Newbold (1974) showed this rejects "no relationship" about **three-quarters** of the time at the 5% level. The regression is not finding a relationship — it is finding that both series wander. **Difference first.**

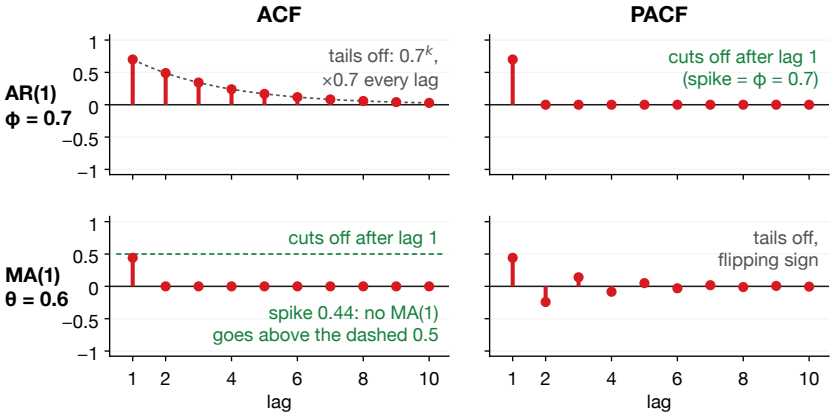
ACF and PACF: Model Identification

The **autocorrelation function**, $\rho_k = \mathrm{Corr}(y_t, y_{t-k})$, measures how y_t relates to its own past. The **partial** autocorrelation controls for the lags in between, which separates a direct effect from one inherited through them.

Read together, they identify the model:

Model	**ACF**	**PACF**
AR(p)	Tails off geometrically	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off geometrically
ARMA	Tails off	Tails off
RW / I(1)	Decays slowly	Big spike at lag 1

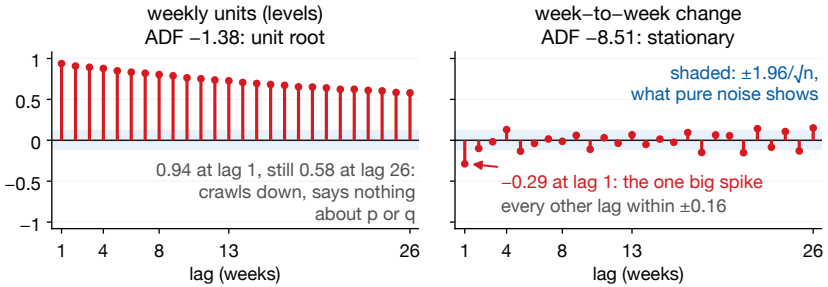
What Those Patterns Actually Look Like



Exact values, not a sample. Read the pair: the sharp cut-off gives the order, the *slow decay* says which side it is on. The AR(1) ACF is a staircase: each lag keeps ϕ of the one before.

Difference First, Then Read the ACF

Lecture 1's TX_1 HOUSEHOLD series, before and after one difference:



The levels' ACF crawls down whatever the process is: it only says you have not differenced yet. The change has one negative spike at lag 1, the MA(1) signature. So ARIMA(0,1,1), which is simple exponential smoothing: Lecture 1's $\alpha = 0.62$ implies $\rho_1 = -0.33$; the data show -0.29.

ARIMA Models

Combine differencing with AR and MA components into one unified model.

ARIMA(p,d,q)

ARIMA(p,d,q) Model

Difference the series d times, $y'_t = \Delta^d y_t$, then fit an ARMA(p,q) to y'_t:

$$y'_t = c + \sum_{i=1}^{p} \phi_i \, y'_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \, \varepsilon_{t-j}$$

Familiar models are corners of this space: **ARIMA(0,1,0)** is a random walk, **ARIMA(1,0,0)** an AR(1), and **ARIMA(0,1,1)** simple exponential smoothing.

A forecast by hand. ARIMA(1,1,0) with $\hat{\phi}_1 = 0.5$, $y_{100} = 120$ and $y_{99} = 118$. The last change was 2, so the next is half of it: $\hat{y}_{101|100} = 120 + 0.5 \times 2 = \mathbf{121}$.

Python: `pmdarima.auto_arima()` searches orders; `statsmodels.SARIMAX` fits the order you choose.

SARIMA: Adding Seasonal Terms

Seasonal data also correlate at lags $m, 2m, 3m, \ldots$, and SARIMA adds AR and MA terms there: **two ARIMA models at different time scales**, (p, d, q) for the short run and $(P, D, Q)[m]$ year over year.

Practical workflow:

1. ADF / KPSS $\rightarrow$ choose d
2. Seasonal unit root test $\rightarrow$ choose D
3. ACF / PACF on the differenced series $\rightarrow$ guide p, q, P, Q
4. Minimize AIC, then check the residual ACF with a Ljung-Box test

Monthly retail: start from SARIMA(1,1,1)(0,1,1)[12] and make anything fancier beat it.

VAR Models

When two series influence each other, model them together symmetrically.

Why Use Multiple Series?

Every model so far — ETS, ARIMA — uses only y_t's own history. But business variables are linked: consumer sentiment today moves retail spending next month, and advertising moves sales over the following weeks.

> If x_t carries information about y_{t+h} that is not already in y_t's own history, including x_t reduces forecast error.

Granger causality (Granger 1969) is the formal test of exactly that claim. It comes right after the VAR.

VAR or ARIMAX: Which Way Does Influence Run?

The modeling choice turns on whether the influence runs one way or both:

VAR — symmetric

- Every variable predicts every other
- Use when influence runs both ways
- No variable is privileged
- Example: retail sales and unemployment (the lab)

ARIMAX — one direction

- x_t drives y_t, not the reverse
- Use when the causal story is clear
- Needs a forecast of x_t itself
- Example: advertising $\rightarrow$ sales

The VAR(p) Model

For two stationary series, each equation regresses on the lags of *both*:

$$\begin{cases} y_{1t} = c_1 + a_{11}\, y_{1,t-1} + a_{12}\, y_{2,t-1} + \varepsilon_{1t} \\ y_{2t} = c_2 + a_{21}\, y_{1,t-1} + a_{22}\, y_{2,t-1} + \varepsilon_{2t} \end{cases}$$

The cross terms carry the value: a_{12} is how much $y_{2,t-1}$ predicts y_{1t}, exactly what a univariate AR throws away. If $a_{12} = a_{21} = 0$, the system is two separate ARs.

Parameter explosion. k variables and p lags make $k^2 p$ coefficients: at $k = 5$, $p = 4$, **100**, often more than a monthly series has observations. Keep $k \leq 4$, or use a LASSO-VAR.

Python: `statsmodels.tsa.api.VAR`; `var.fit(maxlags=p, ic='bic')` picks p by BIC.

Granger Causality Test

Granger Causality

x_t **Granger-causes** y_t if past values of x_t help predict y_t *beyond what* y_t*'s own past already explains.*

The test: fit the y-equation with and without the x-lags; an F-test of H_0: all x-lag coefficients are zero.

In the lab: does unemployment Granger-cause retail sales? It fires *both* ways, and which direction looks stronger depends on whether COVID is in the sample.

Not structural causality. It tests predictive content: a rooster reliably predicts sunrise. Say "predictive of," never "causes."

ARIMAX: Exogenous Variables with ARIMA Errors

When one variable clearly *drives* another, use ARIMAX.

ARIMAX Model

$$y_t = \mathbf{x}_t'\boldsymbol{\beta} + \eta_t, \qquad \eta_t \sim \text{ARIMA}(p, d, q)$$

$\mathbf{x}_t$ are exogenous regressors; η_t absorbs the autocorrelation left in the residuals.

Use it when the direction is clear (advertising $\rightarrow$ sales), you have x_t or can forecast it, and adding it cuts cross-validated error. Check that last one.

The catch: forecasting y_{t+h} needs x_{t+h}, a forecast of your own predictor. Without one, use a VAR, which forecasts every series it contains.

Python: `SARIMAX(y, exog=X, order=(p,d,q)).fit()`

Key Takeaways and Roadmap

Stationarity, ARIMA, VAR, and ARIMAX form the classical time series toolkit.

Key Takeaways

1. **Stationarity first:** test with ADF and KPSS (opposite nulls); difference away a unit root.
2. **ACF** gives the MA order, **PACF** the AR order — read after differencing.
3. **ARIMA(p,d,q)** is an ARMA on the differenced series; SARIMA adds seasonal terms; `auto_arima` searches.
4. **VAR** treats series symmetrically; keep $k \leq 4$. **Granger** tests predictive content, not causation.
5. **ARIMAX** adds exogenous predictors, if you can forecast them.

Next: Generalized Additive Models (GAMs) — flexible, interpretable nonlinear forecasting with Prophet and pyGAM.

References I


Granger, C. W. J. (1969). "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods". In: *Econometrica* 37.3, pp. 424–438.


Granger, C. W. J. and Paul Newbold (1974). "Spurious Regressions in Econometrics". In: *Journal of Econometrics* 2.2, pp. 111–120.